

SFQ-Det: A Structure–Frequency Coupled and Quality-Aware Detector for SAR Ship Detection

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Abstract—Synthetic aperture radar (SAR) imagery enables all-weather maritime observation, but ship detection remains challenging due to speckle noise, complex sea clutter, and small elongated targets. This paper proposes SFQ-Det, a structure–frequency coupled and quality-aware SAR ship detector. SFQ-Det enhances directional ship structures via speckle-aware structural modulation, improves multi-scale robustness through frequency-decoupled wavelet pooling, and introduces a lightweight quality-aware detection head to align classification confidence with localization accuracy. By jointly modeling spatial structure and frequency characteristics, SFQ-Det effectively mitigates speckle interference and improves detection reliability under complex maritime conditions. Experiments on the SSDD dataset demonstrate that SFQ-Det outperforms the YOLO11n baseline while reducing parameters and computation by approximately 36% and 38%, respectively.

Keywords—Synthetic aperture radar, ship detection, structure–frequency coupling.

I. INTRODUCTION

Synthetic aperture radar (SAR) imaging enables all-day and all-weather maritime observation and is widely used in surveillance and reconnaissance. As a key task in marine remote sensing, SAR ship detection has attracted increasing attention, but remains challenging due to severe speckle noise, complex sea clutter, and the small, elongated structure of ship targets [1].

With the development of deep learning, convolutional neural network (CNN)-based detectors, especially one-stage models, have been introduced into SAR ship detection for their efficiency. However, most existing detectors are designed for optical imagery and fail to accommodate the statistical characteristics of SAR data [2][3][4]. Convolution operations tend to confuse speckle noise with structural details, conventional downsampling ignores frequency-domain differences, and classification confidence is usually decoupled from localization accuracy, leading to unreliable detection results in SAR scenarios [5].

Although deep learning–based object detectors, such as YOLO [6], Faster R-CNN [7], and SSD [8], have demonstrated strong performance on standard benchmarks, they still encounter substantial difficulties in complex SAR image

scenarios. In particular, their detection accuracy often degrades under severe speckle noise and cluttered backgrounds, where discriminative target features are easily overwhelmed by noise-induced responses.

Several studies have attempted to address these issues from different perspectives. Cui et al. [9] proposed MSF-SAR by enhancing contextual modeling and multi-scale feature fusion for small-ship detection. Hu et al. [10] introduced SDWPNet, which incorporates wavelet-based downsampling to preserve critical details under lightweight constraints. Fan et al. [11] proposed a hybrid logarithmic sparse representation model to improve robustness in small-sample scenarios. Despite these advances, how to jointly handle speckle-dominated backgrounds and multi-scale variations under strict computational constraints remains an open problem.

To this end, this paper proposes SFQ-Det, a structure–frequency coupled and quality-aware SAR ship detector. A speckle-aware guided structure modulation mechanism enhances directional ship structures while suppressing pseudo high-frequency responses. A speckle-aware wavelet decomposition pooling operator enables frequency decoupling and adaptive suppression during scale transformation. In addition, a shared lightweight quality-aware detection head explicitly integrates localization quality into classification confidence. Experiments on the SSDD [12] dataset demonstrate that SFQ-Det outperforms the YOLO11n baseline with significantly reduced model complexity.

II. PROPOSED METHOD

A. Overall Architecture of SFQ-Det

This section provides an overview of SFQ-Det, including the speckle-aware guided structure modulation module, the speckle-aware wavelet decomposition pooling operator, and the shared lightweight quality-aware detection head. The overall architecture is illustrated in Fig. 1. SFQ-Det is built upon the one-stage YOLO11 framework with targeted architectural modifications, where each component is carefully designed to enhance structural representation, frequency robustness, and prediction reliability in complex SAR scenes.

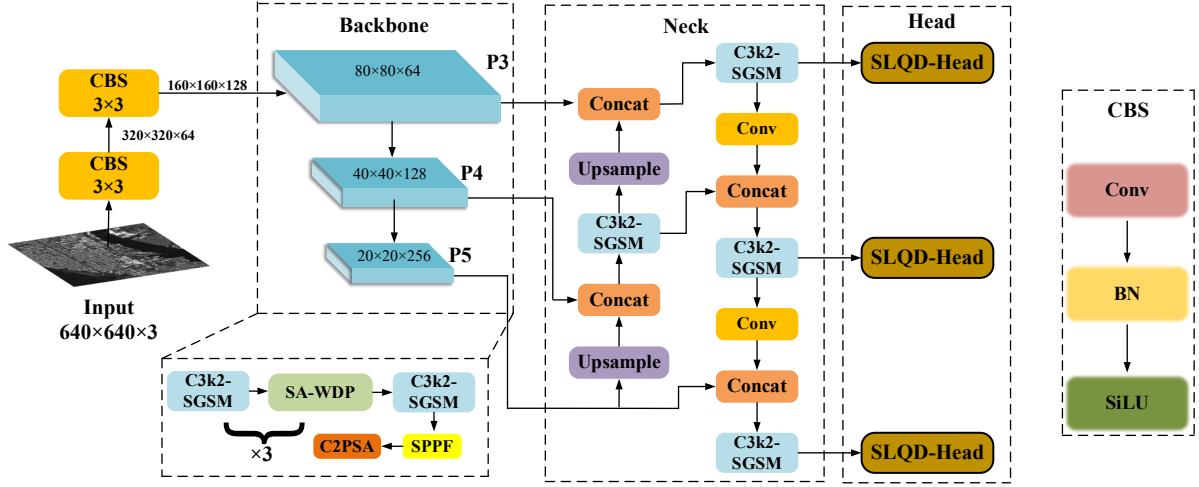


Fig. 1. Overall architecture of the proposed SFQ-Det. The network is built upon the YOLO11n baseline, with SGSM and SA-WDP integrated into the backbone and a shared lightweight quality-aware detection head (SLQD) for improved SAR ship detection.

B. Speckle-aware Guided Structure Modulation

Due to the coherent imaging mechanism, SAR images suffer from strong speckle noise and unstable local statistics, leading to noise-dominated high-frequency responses that make conventional convolution operators prone to confusing true structural information with speckle-induced pseudo responses. To address this issue, we propose a Speckle-aware Guided Structure Modulation (SGSM) module (Fig. 2), which enhances structural representations by jointly modeling local consistency, directional patterns, and global responses through lightweight structural modeling and statistical modulation.

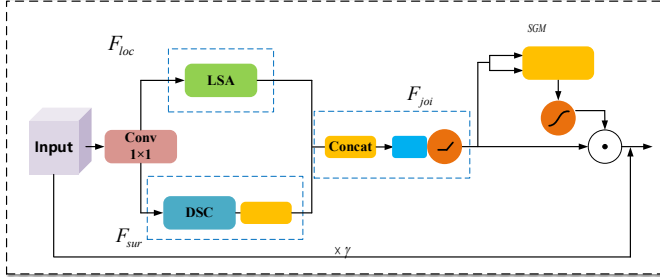


Fig. 2. Architecture of the speckle-aware guided structure modulation (SGSM) module.

Given an input feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, SGSM first applies a 1×1 convolution for channel compression, producing $\mathbf{X}' \in \mathbb{R}^{\frac{C}{2} \times H \times W}$, which is then fed into two parallel structural modeling branches.

In the local branch, Local Shift Aggregation (LSA) aggregates the center position and its four neighbors:

$$\mathbf{F}_{loc} = \frac{1}{5}(\mathbf{X}' + S_{+1,0}(\mathbf{X}') + S_{-1,0}(\mathbf{X}') + S_{0,+1}(\mathbf{X}') + S_{0,-1}(\mathbf{X}')) \quad (1)$$

which stabilizes local structural responses by averaging random speckle fluctuations.

The directional branch adopts Directional Structure Contrast (DSC) to enhance direction-consistent ship and wake patterns:

$$\mathbf{F}_{sur} = \frac{1}{2}(S_{+,d,0}(\mathbf{X}') - S_{-,d,0}(\mathbf{X}') + S_{0,+,d}(\mathbf{X}') - S_{0,-,d}(\mathbf{X}')). \quad (2)$$

To suppress speckle-dominated abnormal responses, a Speckle Energy Gate (SEG) is applied based on the global energy of directional features:

$$g_{spk} = \sigma\left(-\alpha \frac{1}{CHW} \sum_{c,h,w} |\mathbf{F}_{sur}(c,h,w)| + \beta\right), \quad (3)$$

yielding the gated directional feature:

$$\tilde{\mathbf{F}}_{sur} = g_{spk} \cdot \mathbf{F}_{sur}. \quad (4)$$

The local and directional features are fused and further refined by a lightweight Statistical Global Modulation (SGM) module, producing the final output via a scaled residual connection:

$$\mathbf{Y} = \mathbf{X} + \gamma \cdot \mathbf{F}_{out}, \quad (5)$$

where γ is initialized to zero and gradually learned during training.

Through these designs, SGSM effectively enhances direction-consistent ship structures while suppressing speckle-induced pseudo responses, providing robust structural features for subsequent frequency-domain modeling and detection.

C. Speckle-aware Wavelet Decomposition Pooling

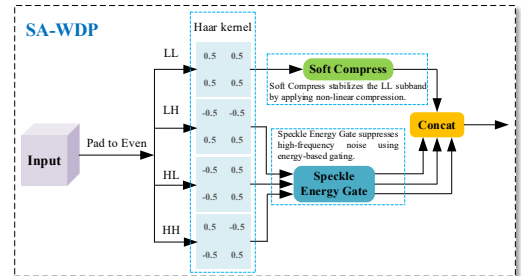


Fig. 3. Architecture of the speckle-aware wavelet decomposition pooling (SA-WDP) module.

In SAR ship detection, conventional strided downsampling mixes frequency components, amplifying speckle-induced high-frequency noise and degrading small-target discrimination. To address this issue, we propose a Speckle-Aware Wavelet Decomposition Pooling (SA-WDP) module (Fig. 3) that performs frequency-aware downsampling via wavelet-based decoupling and speckle-adaptive suppression [13].

Given an input feature map $\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$, SA-WDP applies Haar wavelet decomposition to obtain one low-frequency subband and three high-frequency subbands:

$$\{\mathbf{LL}, \mathbf{LH}, \mathbf{HL}, \mathbf{HH}\} = \mathcal{W}(\mathbf{X}), \quad (6)$$

where LL captures coarse structural information, while LH, HL, and HH encode directional high-frequency details.

To stabilize low-frequency responses with large dynamic ranges, a lightweight soft compression is applied:

$$\widetilde{\mathbf{LL}} = \frac{\mathbf{LL}}{1 + |\mathbf{LL}| + \varepsilon}, \quad (7)$$

where ε is a small constant for numerical stability.

For high-frequency subbands dominated by speckle noise, a Speckle Energy Gate (SEG) is introduced to adaptively suppress noise:

$$g_k = \sigma(-a_k e_k + b_k), \quad \tilde{\mathbf{F}}_k = g_k \cdot \mathbf{F}_k, \quad (8)$$

After low-frequency compression and high-frequency gating, SA-WDP concatenates the processed subbands along the channel dimension to produce the final output feature:

$$\mathbf{Y} = \text{Concat}[\widetilde{\mathbf{LL}}, \widetilde{\mathbf{LH}}, \widetilde{\mathbf{HL}}, \widetilde{\mathbf{HH}}] \in \mathbb{R}^{B \times 4C \times \frac{H}{2} \times \frac{W}{2}}. \quad (9)$$

where e_k denotes the global energy of subband $k \in \{\mathbf{LH}, \mathbf{HL}, \mathbf{HH}\}$.

Through these designs, SA-WDP enables frequency-aware downsampling with speckle-adaptive suppression, providing robust multi-scale features for subsequent detection. Together with SGSM, it complements spatial structural modeling with frequency-domain robustness under complex SAR sea clutter.

D. Shared Lightweight Quality-aware Detection Head

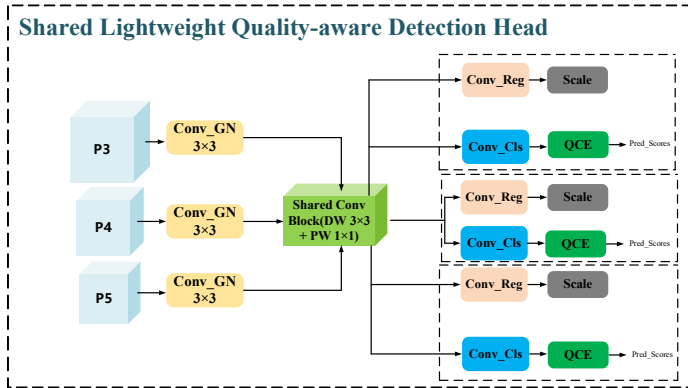


Fig. 4. Architecture of the shared lightweight quality-aware detection head (SLQD).

In SAR ship detection, speckle noise and sea clutter often lead to misaligned classification confidence and localization accuracy, especially in lightweight detectors. To address this issue, we propose a Shared Lightweight Quality-aware Detection Head (SLQD) that enhances prediction reliability via cross-scale parameter sharing and localization-quality-guided classification while preserving detection efficiency [14].

Given the multi-scale feature set from the backbone, $\mathcal{P} = \{P_3, P_4, P_5\}$, each feature is first processed by an independent $3 \times 3 \text{Conv-GN}$ module for channel alignment:

$$\mathbf{F}_i = \text{Conv}_{\text{GN}}^{3 \times 3}(P_i), \quad i \in \{3, 4, 5\}. \quad (10)$$

The aligned features are then fed into a shared depthwise separable convolution module:

$$\tilde{\mathbf{F}}_i = \text{Conv}_{\text{PW}}^{1 \times 1}(\text{Conv}_{\text{DW}}^{3 \times 3}(\mathbf{F}_i)), \quad (11)$$

enabling consistent modeling across scales while reducing parameters and computation.

For prediction, the shared features are sent to both regression and classification branches. The regression branch predicts bounding box distributions with a scale-adaptive factor:

$$\mathbf{B}_i = s_i \cdot \text{Conv}_{\text{reg}}(\tilde{\mathbf{F}}_i), \quad (12)$$

To explicitly align classification confidence with localization quality, a Quality-aware Classification Estimator (QCE) is introduced. A localization quality descriptor is extracted from regression outputs:

$$\mathbf{Q}_i = \phi(\mathbf{B}_i), \quad (13)$$

and used to calibrate classification confidence:

$$\hat{\mathbf{C}}_i = \sigma(\mathbf{C}_i \odot \psi(\mathbf{Q}_i)). \quad (14)$$

This design suppresses overconfident predictions caused by low-quality localization.

Finally, SLQD concatenates $\mathbf{B}_i$ and $\hat{\mathbf{C}}_i$ to form multi-scale detection outputs. Through these designs, SLQD achieves lightweight cross-scale parameter sharing and explicit classification-localization consistency, significantly improving detection stability in complex SAR scenarios.

III. EXPERIMENTS

A. Experimental Settings

Experiments were conducted on a Windows 10 platform with an NVIDIA RTX 2060 GPU (6 GB) and an AMD Ryzen 7 4800H CPU (2.90 GHz), using PyTorch 2.5.1 (CUDA 12.6) and Python 3.9. Evaluations were performed on the SSDD dataset (1,160 SAR images with 2,456 ship annotations), which was randomly split into training, validation, and test sets at a ratio of 70%/10%/20%, and performance was evaluated using mAP@0.5, Recall, GFLOPs, and the number of parameters.

All images were resized to 640×640. The model was trained for 350 epochs with a batch size of 16 using the SGD optimizer,

with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005.

B. Ablation Experiments

TABLE I
RESULTS OF ABLATION EXPERIMENTS ON THE SSDD DATASET

Models	SGSM	SA-WDP	SLQD-Head	Params(M)	FLOPs(G)	mAP@0.5	R%
A				2.58	6.3	96.1	89.3
B	✓			2.18	5.5	97.3	92.4
C		✓		2.23	5.5	97.5	91.3
D			✓	2.43	5.7	97.3	92.6
E	✓	✓	✓	1.66	3.9	98.2	95.6

Ablation experiments were conducted on the SSDD dataset to evaluate the contribution of each module, as summarized in Table I. Model A denotes the baseline, Models B–D individually incorporate SGSM, SA-WDP, and SLQD-Head, and Model E represents the full model integrating all modules.

As shown in Table I, each module yields consistent performance gains when applied independently, validating the effectiveness of structure-aware modeling, frequency-domain decoupling, and quality-aware detection. Notably, the improvements brought by different modules are complementary rather than redundant, indicating that each module addresses distinct challenges in SAR ship detection. The full model achieves the best performance, attaining 98.2% mAP@0.5 and 95.6% Recall with only 1.66 M parameters and 3.9 GFLOPs, demonstrating a strong accuracy–efficiency trade-off.

C. Comparison Experiments

TABLE II
THE COMPARATIVE EXPERIMENT ON THE SSDD DATASET.

Models	Params(M)	FLOPs(G)	mAP@0.5	R%
YOLOv5m	20.85	48.2	94.5	84.9
YOLOv8	3.01	8.2	95.7	92.0
YOLO11n	2.58	6.3	96.8	93.5
CSFF-MGDH	-	-	96.7	-
Improved YOLOv7	-	-	96.5	90.3
LD-YOLO	4.5	12.0	98.2	95.0
SFQ-Det	1.66	3.9	98.2	95.6

To evaluate the overall performance of SFQ-Det, comparative experiments were conducted on the SSDD dataset, as shown in Table II. YOLOv5m provides strong representation capability but incurs high computational cost, while lightweight variants such as YOLOv8 and YOLO11n reduce model complexity yet remain challenged by complex sea clutter.

For SAR-specific baselines, CSFF-MGDH [15] enhances the YOLOX framework by introducing deformable convolution and cross-stage feature fusion to better model complex backgrounds and multi-scale targets. Improved YOLOv7 [16] employs dual-channel pooling attention and spatial weight balancing to strengthen multi-scale feature alignment. LD-YOLO [17], built upon YOLOv8n, integrates dynamic convolution, large-kernel attention, and a dedicated detection head to improve small-target detection. In contrast, SFQ-Det achieves 98.2% mAP@0.5 and 95.6% Recall with only 1.66 M parameters and 3.9 GFLOPs, matching the accuracy of LD-YOLO while significantly reducing model complexity, indicating a more favorable accuracy–efficiency trade-off.

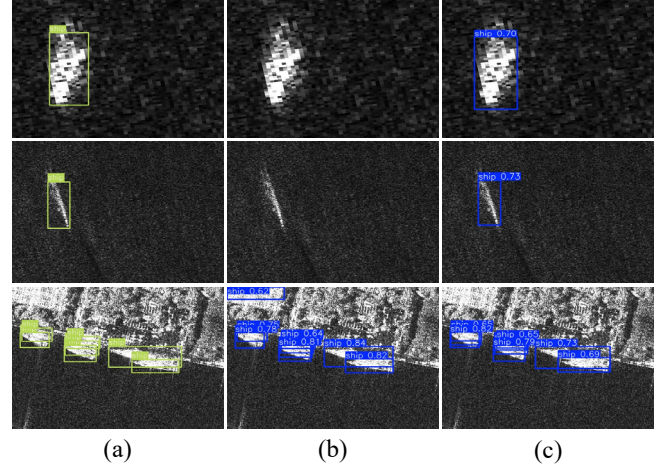


Fig. 5. Visual comparison of SAR ship detection on the SSDD dataset.

Fig. 5 presents qualitative detection results on the SSDD dataset, where (a) shows the ground truth, (b) the baseline YOLO11n, and (c) the proposed SFQ-Det. As illustrated, YOLO11n tends to miss small and elongated ships or produce inaccurate localization under complex sea clutter. In contrast, SFQ-Det yields detection results that are more consistent with the ground truth, exhibiting improved localization accuracy and target separation, particularly in dense multi-ship scenes. These results demonstrate the robustness and effectiveness of SFQ-Det in complex SAR environments.

IV. CONCLUSION

This paper presents SFQ-Det, a lightweight and robust SAR ship detection framework. By integrating structure–frequency coupled modeling and quality-aware detection, SFQ-Det effectively addresses challenges such as complex sea clutter, small targets, and elongated ship structures. Experiments on the SSDD dataset demonstrate that SFQ-Det achieves high detection accuracy with significantly reduced computational cost, offering a favorable accuracy–efficiency trade-off. These results indicate that SFQ-Det is well suited for practical deployment in resource-constrained SAR ship detection scenarios. Future work will focus on further improving real-time performance and generalization in more complex maritime environments.

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